

Carcinogenesis in the GI tract: from morphology to genetics and back again.

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The genetic alterations in colorectal cancer progression are determined by one of two separate and distinct underlying pathways of genomic instability. The first pathway, chromosomal instability, is characterized by allelic losses and aneuploidy. The second pathway, microsatellite instability, is characterized by an abundance of subtle DNA mutations and diploidy. Although the genes causing chromosomal instability remain unknown, microsatellite instability is caused by inactivation of a DNA mismatch repair gene (predominantly MLH1 or MSH2). Microsatellite instability is present in 15% of colorectal cancers, and is diagnosed by analysis of tumor DNA from paraffin blocks and by demonstration of loss of mismatch repair protein expression in cancers. In addition to the unique profile of genetic alterations, colorectal cancers with microsatellite instability have distinct pathologic features and improved survival. Finally, cancers from most patients with hereditary non-polyposis colorectal cancer (or Lynch syndrome) have microsatellite instability due to germline mutations in the DNA mismatch repair genes. Identification of the microsatellite instability pathway has enormous implications for the clinical investigation and management of colorectal cancer patients.